AQA

# FUNCTIONAL SKILLS CERTIFICATE

# Functional Mathematics

Level 1

Mark Scheme

4367

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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

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**Glossary for Mark Schemes**

Examinations are marked to award positive achievement.

Marks are awarded for demonstrating the following interrelated **process skills**.

**Representing** Selecting the mathematics and information to model a situation.

**R.1** Candidates recognise that a situation has aspects that can be represented using mathematics.

**R.2** Candidates make an initial model of a situation using suitable forms of representation.

**R.3** Candidates decide on the methods, operations and tools, including ICT, to use in a situation.

**R.4** Candidates select the mathematical information to use.

**Analysing** Processing and using mathematics.

**A.1** Candidates use appropriate mathematical procedures.

**A.2** Candidates examine patterns and relationships.

**A.3** Candidates change values and assumptions or adjust relationships to see the effects on answers in models.

**A.4** Candidates find results and solutions.

**Interpreting** Interpreting and communicating the results of the analysis.

**I.1** Candidates interpret results and solutions.

**I.2** Candidates draw conclusions in light of situations.

**I.3** Candidates consider the appropriateness and accuracy of results and conclusions.

**I.4** Candidates choose appropriate language and forms of presentation to communicate results and solutions.

In particular, individual marks are mapped onto the following **skills standards**.

**Representing** Making sense of the situations and representing them.
A learner can:

**Ra** Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.

**Rb** Identify the situation or problems and identify the mathematical methods needed to solve them.

**Rc** Choose from a range of mathematics to find solutions.

**Analysing** Processing and using the mathematics.
A learner can:

**Aa** Apply a range of mathematics to find solutions.

**Ab** Use appropriate checking procedures and evaluate their effectiveness at each stage.

**Interpreting** Interpreting and communicating the results of the analysis.
A learner can:

**Ia** Interpret and communicate solutions to multistage practical problems in familiar and unfamiliar contexts and situations.

**Ib** Draw conclusions and provide mathematical justifications.

To facilitate marking, the following categories are used:

**M** Method marks are awarded for a correct method which could lead to a correct answer.

**A** Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.

**B** Marks awarded independent of method.

**ft** Follow through marks. Marks awarded following a mistake in an earlier step.

**SC** Special case. Marks awarded within the scheme for a common misinterpretation which has some mathematical worth.

**oe** Or equivalent. Accept answers that are equivalent.
eg, accept 0.5 as well as $\frac{1}{2}$

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **1(a)** | 2 × 5.79 + 2 × 4.79<br>or<br>11.58 + 9.58<br>or<br>21.16 | M1<br>*Rb* | |
| | their 21.16 – 18.99<br>or<br>18.99 + 3 or 21.99<br>or<br>21.16 – 3 or 18.16 | M1<br>*Aa* | their 21.16 must be from an attempt at the cost of at least 2 tickets<br>the subtraction may then be reversed |
| | 2.17 and No<br>or<br>21.16 and 21.99 and No<br>or<br>18.16 and No | A2<br>*I* | A1 2.17<br>or<br>A1 21.16 and 21.99<br>or<br>A1 18.16<br>A1 ft correct decision for their value(s) if 2nd M1 awarded and cost is attempted for exactly 4 tickets<br><br>SC2 21.16 and No |
| | **Additional Guidance** | | |
| | No can be implied eg Its less than £3<br>Example of ft<br>4 x 5.79 = 23.16<br>23.16 – 18.99 = 4.17 Yes M0M1A0A1ft | | |

| | | | |
|---|---|---|---|
| **1(b)** | 8 | B1<br>*Aa* | |
| | 17 | B1ft<br>*Aa* | correct total of their 8 + 9 |
| | **Additional Guidance** | | |
| | If frame 1 is blank award B1 for 17 in frame 2 | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **1(c)** | (Frame 9=) 75 + 10 + 3 + 5 or 93 | M1<br>*Ra* | 93 may be written in frame 9 total |
| | their 93 + 3 + 5 | M1<br>*Aa* | their 93 must be >75<br>75 + 10 + 3 + 5 + 3 +5<br>or<br>75 + 26 implies M2 |
| | 101 and Yes | A2<br>*I* | A1 101<br>A1 ft correct decision for their value if 2nd M1 awarded<br>SC2 88 and 91 and No<br>SC1 88 and 91 |
| | **Additional Guidance** | | |
| | Working may be seen on the scorecard.<br>Mark working lines if different to scorecard.<br>85 in frame 9 and 93 in frame 10 and No scores M0M1A0A1ft<br>Final answer of 93 and No gains M0M1A0A1ft<br>This may be in the working lines. | | |

| | | | |
|---|---|---|---|
| **1(d)** | Jamil/He won 3 games (out of 5) but Tom (only) won 2 (out of 5)<br>or<br>Jamil/He won a greater proportion of games/more games (than Tom)<br>or<br>Jamil/He won 60% of the games but Tom (only) won 40% | B1<br>*I* | oe<br>eg Tom lost 3 games but Jamil (only) lost 2 |
| | **Additional Guidance** | | |
| | Ignore other non-contradictory working or comments unless they state that Jamil is not correct/supported. | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 1(e) | **Alternative method 1** | | |
| | **Compares totals**<br>145 +138 + 204 + 186 + 172<br>or<br>192 + 165 + 144 + 210 + 184 | M1<br>*Rc* | |
| | 845 and 895 | A2<br>*Aa* | A1 845 or 895 |
| | **Alternative method 2** | | |
| | **Compares means**<br>(145 +138 + 204 + 186 + 172) ÷ 5<br>or 845 ÷ 5<br>or<br>(192 + 165 + 144 + 210 + 184) ÷ 5<br>or 895 ÷ 5 | M1<br>*Rc* | |
| | 169 and 179 | A2<br>*Aa* | A1 169 or 179 |
| | **Alternative method 3** | | |
| | **Compares medians**<br>138, 145, 172, 186, 204<br>or<br>144, 165, 184, 192, 210 | M1<br>*Rc* | ordering |
| | 172 and 184 | A2<br>*Aa* | A1 172 or 184<br>Must be evidence that they are working out median. Eg states median or shows method |
| | **Alternative method 4** | | |
| | 192 – 145, 165 – 138, 144 – 204,<br>210 – 186 and 184 – 172<br>or<br>145 – 192, 138 – 165, 204 – 144,<br>186 – 210 and 172 – 184 | M1<br>*Rc* | |
| | Tom (+)47, (+)27, (-)60, (+)24, (+)12<br>or<br>Jamil (-) 47, (-) 27, (+) 60 (-)24, (-) 12 | A1<br>*Aa* | |
| | Tom +50<br>or Jamil -50 | A1<br>*Aa* | Allow clear comparison of difference totals<br>eg 47 + 27 + 24 + 12 is clearly more than Jamil's +60 |
| | **Additional Guidance** | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| | Evidence of method may be seen under/next to tables.<br>For Alt 4 other ways of comparing the differences may be seen.<br>eg Toms 4 positive differences listed and Jamil's 1 positive difference.<br>Ignore other non-contradictory working or comments unless they state that Tom is not correct/supported. | | |

| | | | |
|---|---|---|---|
| **2(a)** | Tuna | B1<br>*Aa* | |
| | **Additional Guidance** | | |
| | | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **2(b)** | **Alternative method 1** | | |
| | (with salad)<br>29 + 20 + 32 + 25 or 106<br>or<br>(without salad)<br>26 + 30 + 36 + 22 or 114 | M1<br>*Rb* | Allow one error when reading from graph |
| | 106 and 114 and No | A2<br>*I* | A1 106 and 114<br>A1ft Correct decision for their values |
| | **Alternative method 2** | | |
| | Differences<br>(without salad)<br>-3, +10, +4, -3<br>or<br>(with salad)<br>+3, -10, -4, +3 | M1<br>*Rb* | Allow one error |
| | +14 and -6 and No<br>or<br>-14 and +6 and No<br>or<br>8 and No<br>or<br>8 more without salad<br>or<br>8 less with salad | A2<br>*I* | A1 +14 and -6<br>or -14 and +6<br>or 8<br>A1ft Correct decision for their value(s) |
| | **Additional Guidance** | | |
| | M1 can be awarded for indicating addition of 4 corresponding values with at most one error.<br>Example<br>27 + 20 + 32 + 25 and 26 + 30 + 34 + 23 gains M1 for the first 4 (one error) even though the 2nd list has two errors.<br>Allow 'they sold more without salad' to imply 'No'<br>May need to look at graph | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **2(c)** | 300 ÷ 25 | M1<br>*Rc* | |
| | 12 | A1<br>*Aa* | |
| **2(c) check** | Reverse method<br>eg 12 × 25 = 300<br>or 300 ÷ 12 = 25<br>or<br>Alternative method<br>eg 300 ÷ 5 = 60 and 60 ÷ 5 = 12 | B1<br>*Ab* | |
| | **Additional Guidance** | | |
| | Mark holistically. | | |

| | | | |
|---|---|---|---|
| **2(d)** | **Alternative method 1** | | |
| | 2 × 1000 or 2000 | M1<br>*Aa* | |
| | 500 × their 12 or 6000 | M1<br>*Ra* | ft their 12 from **(c)** |
| | their 6000 ÷ their 2000 | M1<br>*Rc* | their total grams needed ÷ their grams in 2kg |
| | 3 | A1ft<br>*Aa* | only ft their 12 from **(c)** |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 2(d) | **Alternative method 2** | | |
| | 500 × their 12<br>or<br>$\frac{500}{25} \times 300$<br>or 6000 | M1<br>*Aa* | ft their 12 from **(c)**<br><br>Allow complete build up method to 500<br>eg<br>25 = 300(g)<br>50 = 600(g)<br>500 = 6000(g) |
| | their 6000 ÷ 1000 or 6 | M1<br>Ra | their total grams needed ÷ 1000 |
| | their 6 ÷ 2 | M1<br>Rc | their 6 must come from converting their grams to kg |
| | 3 | A1ft<br>*Aa* | only ft their 12 from **(c)** |
| | **Alternative method 3** | | |
| | 2 × 1000 or 2000 | M1<br>*Aa* | |
| | their 2000 ÷ 300 × 25<br>or<br>their 2000 ÷ their 12<br>or 166.(66…) or 166.7 | M1<br>Ra | |
| | 500 ÷ their 166.(66….) | M1<br>Rc | |
| | 3 | A1ft<br>*Aa* | only ft their 12 from **(c)** |
| | **Additional Guidance** | | |
| | Allow 3 with no working for full marks.<br>If their 12 from **c** leads to a decimal accept the complete decimal or rounding up correctly to any sf. | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 2(e) | **Alternative method 1** | | |
| | 90 ÷ 30 or 3<br>or<br>140 ÷ 60 or 2.3(..) or 2<br>or<br>90 ÷ 60 or 1.5 or 1<br>or<br>140 ÷ 30 or 4.6(..) or 4.7 or 4 | M1<br>*Ra* | |
| | their 3 × their 2 | M1<br>*I* | Any decimal value(s) must be rounded down to integer(s)<br>Multiplication can be implied by answer |
| | 6 | A1<br>*Aa* | from correct working<br>SC2 6 from no working or insufficient correct working |
| | **Alternative method 2** | | |
| | Draws one correct crate | M1<br>*Ra* | |
| | Draws at least 4 correct size crates | M1<br>*I* | ignore additional incorrect sized crates |
| | 6 | A1<br>*Aa* | All 6 crates of correct size clearly shown<br>SC2 6 from no working or insufficient, correct working |
| | **Additional Guidance** | | |
| | Mark the method that leads to the answer stated. If no answer is given mark the best method.<br>If the answer space is blank then 6 correctly drawn boxes numbered to 6 gains M2A1<br>Allow dots to indicate the corners of boxes. | | |
| | Use of area divided by area given as their answer (7.(875)) is M0M0A0 Ignore any diagram. | | |
| | For SC2 the insufficient, correct working is most likely to come from the diagram method.<br>These may include unclear lines drawn for some boxes or not all 6 boxes shown.<br>To merit 3 marks the 6 boxes must be clearly shown and either numbered to 6 or 6 stated. | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **2(f)** | Fully communicated shortest route chosen starting shop to A and 8 (miles)<br>(Shop) → A → C → B → Shop and 8 (miles) | B3<br>*Rb Aa I* | B2 The correct shortest route with incorrect or no distance or not indicated as their choice<br>or<br>Shortest route but starting shop to B and 8 miles<br>(Shop) →B →C →A →Shop and 8 miles<br>or<br>A possible route starting shop to A with correct distance (but not the shortest)<br>eg1 (Shop) →A → B → C → Shop and $10\frac{1}{2}$ (miles)<br>eg2 (Shop) → A → C → Shop → B→ Shop and $8\frac{1}{2}$ (miles)<br><br>B1 A possible route starting shop to A but with incorrect or no total distance<br>eg1 (Shop) →A → B → C → Shop and $11\frac{1}{2}$ (miles)<br>or<br>B1 Route with correct distance visiting all 3 offices but either not starting shop to A or not finishing at the shop<br>eg1 (Shop) → C → B → A → Shop and $10\frac{1}{2}$ (miles)<br>eg2 (Shop) → A → B → C and $7\frac{1}{2}$ (miles)<br>**Note** the exception to this is the shortest route of 8 miles starting shop to B as shown in B2 above |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| | **Additional Guidance** | | |
| | Penalise, by 1 mark, routes not communicated by letters.<br>Examples<br>1½ + 2 + 3½ + 1 = 8 B2 (B3 -1)<br>This is the correct distance for the shortest route (Shop) → A → C → B → Shop<br><br>1 + 2½ + 2 + 3 = 8½ B0 (B1 -1)<br>This is the correct distance for the route (Shop) → B → A → C → Shop<br><br>Omitting the word shop at the end but giving the correct distance including return to shop loses the communication mark.<br><br>If a choice is made, mark that choice.<br>If a choice is not made the maximum mark allowed is B2. Award the best attempt.<br>Eg SABCS = 10½<br>SCBAS = 10½<br>Award B2 for first alternative | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **3(a)** | Two sinks of correct size against North wall | B1<br>*Ra* | |
| | Three circles of approx. correct size in a line in the South half | B1<br>I | May be sketched |
| | One rectangle (for desk) measuring 1 cm by 2 cm | B1<br>Aa | |
| | One rectangle (for waiting area) 4cm by 2cm | B1<br>Aa | |
| | Door area kept clear and all items labelled correctly | B1<br>*I* | |
| | **Additional Guidance** | | |
| | | | |
| **3(b)** | 2 different people on each shift | B1<br>*Ra* | |
| | Jenny, Craig and Mia included 8 times each | B1<br>*Aa* | |
| | Craig does not work on Saturday | B1<br>*I* | |
| | Each person has at least 1 day off | B1<br>*I* | |
| | **Additional Guidance** | | |
| | For 2nd B1 condone twice on the same shift | | |
| | If there are blanks then the only possible mark is the 3rd B1 if Saturday is completed using only Jenny and Mia. | | |
| | Mark 2nd grid unless blank. | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **3(c)** | 12 × 60 or 720<br>or<br>30 × 33 or 990 | M1<br>*Rb* | |
| | 12 × 60 + 30 × 33<br>or<br>their 720 + their 990<br>or 1710 | M1<br>*Rc* | 2 different values<br>their 720 must be >60<br>their 990 must be >33 |
| | 704 + 300 or 1004 | M1<br>*Aa* | |
| | their 1710 – their 1004<br>or<br>their 1004 + 700 or 1704<br>or<br>their 1710 – 700 or 1010 | M1<br>*Rc* | their 1710 – 704 – 300 implies previous M1<br>their 1710 can be 93 (from 60 + 33)<br>their income – both costs |
| | 706 and Yes<br>or<br>1710 and 1704 and Yes<br>or<br>1010 and 1004 and Yes | A2<br>*I* | A1 706<br>or 1710 and 1704<br>or 1010 and 1004<br>A1ft correct decision for their value(s) if 3rd and 4th M1 awarded |
| | **Additional Guidance** | | |
| | | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 4(a) | 3 × 6 or 18 | M1<br>*Ra* | Allow 3000 × 6 |
| | their 18 × 15 (÷ 100)<br>or<br>270 (÷100)<br>or<br>250 ÷ 15<br>or<br>2.5(0) ÷ 0.15 | M1<br>*Aa* | their 18 cannot be 3 or 6 |
| | (£)2.7(0) and Yes<br>or 16.(....) and 18 and Yes | A2<br>*I I* | A1 (£)2.7(0) or 16.(....) and 18<br>A1ft correct decision for their values if M2 awarded |
| | **Additional Guidance** | | |
| | 16.(..) comes from calculating the number of units you can buy for £2.50<br>Allow any rounding including rounding up to 17<br>Answer 270p and Yes scores M2A2<br>Answer 270 and 250 and Yes scores M2A2<br>Answer 270 with no units and Yes scores M2A0A1<br><br>6 × 15 ( = 90) oe scores M0<br><br>Use of 3000 × 6 needs division by 1000 for the accuracy mark<br>Example<br>3000 × 6 = 18000<br>18000 × 0.15 = 270 Yes M2 A0 A1ft<br><br>Use of 2kw or 1kw instead of 3kw clearly shown can gain max 3 marks | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 4(b) | **Alternative method 1** | | |
| | 100 ÷ 1000 or 0.1 or $\frac{1}{10}$ | M1<br>*Ra* | |
| | their 0.1 × 500 or 50<br>or 500 ÷ 10 or 50<br>or 500 × 4 or 2000 | M1<br>*Aa* | |
| | 50 × 4 = 200<br>or 200 ÷ 4 = 50<br>or 2000 × 0.1 = 200 | A1<br>*I* | 50 must be from 0.1 × 500 |
| | **Alternative method 2** | | |
| | 100 × 500 or 50 000 | M1<br>*Ra* | |
| | their 50 000 ÷ 1000 or 50 | M1<br>*Aa* | |
| | 50 × 4 = 200<br>or 200 ÷ 4 = 50 | A1<br>*I* | 50 must be from 50 000 ÷ 1000 |
| | **Alternative method 3** | | |
| | 500 × 4 or 2000 | M1<br>*Ra* | 500 × 100 or 50 000 |
| | their 2000 × 100 or 200 000 | M1<br>*Aa* | their 50 000 × 4 or 200 000 |
| | 200 000 ÷ 1000 = 200<br>or<br>200 × 1000 = 200 000 | A1<br>*I* | |
| | **Additional Guidance** | | |
| | 50 × 4 = 200 or 200 ÷ 4 = 50 on its own is M0M0A0<br>Any 50 used must clearly be from using the method in alt 1 or alt 2 | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| **4(c)** | 200 ÷ 5 | M1<br>*Rc* | |
| | 40 | A1<br>*Aa* | |
| **4(c) check** | Reverse or alt method<br>eg reverse<br>their 40 × 5 = 200<br>or 200 ÷ their 40 = 5<br>eg alt method<br>200 ÷ 10 × 2 = 40<br>or 200 × 0.2 = 40 | B1ft<br>*Ab* | |
| | **Additional Guidance** | | |
| | Mark holistically so two different methods of 200 ÷ 5 = 40 and 200 ÷ 10 × 2 = 40 gains M1A1B1 | | |

| Q | Answer | Mark | Comments |
|---|---|---|---|
| 4(d) | **Alternative method 1** | | |
| | Their 40 × 15 (÷ 100) or 600 or 6 | M1<br>*Ra* | ft from **(c)**<br>implied by 19.88 |
| | 38 – (their 6 + 13.88)<br>or<br>20 + (their 6 + 13.88) | M1<br>*Rc* | For 38 condone 68 |
| | (£)18.12 and No<br>or<br>(£)39.88 and No | A2ft<br>*I I* | ft from **(c)**<br>A1 (£)18.12 or (£)39.88<br>or<br>A1ft correct decision for their value if M2 awarded |
| | **Alternative method 2** | | |
| | their 40 × 15 (÷ 100) or 600 or 6 | M1<br>*Ra* | ft from **(c)** |
| | 38 – 20 or 18<br>and<br>their 6 +13.88 or 19.88 | M1<br>*Rc* | For 38 condone 68 |
| | (£)18 and (£)19.88 and No | A2ft<br>*I I* | ft from **(c)**<br>A1 (£)18 and (£)19.88<br>or<br>A1ft correct decision for their values if M2 awarded |
| | **Additional Guidance** | | |
| | 68 comes from adding the cost of 200 units at 15p to the cost for the ordinary bulbs- a misinterpretation<br>This gives 68 – 19.88 = 48.12 Yes M1M1 A0A1ft<br>Equivalent comparisons should be given full credit<br>Example<br>40 × 0.15 = 6<br>38 – 20 – 6 =12 No M2A2<br>(this is a comparison of £12 with £13.88) | | |